



Compare and revise model plates

Autumn 1 · Week 6 · Lesson B · 40 minutes

Learning objective

Compare two fictional food-card models, replace a protein source and justify the group relationship while stating model limits.

Evidence to look for

Record the pupil decision or explanation and the support used. Accept pointing, signing, adult scribing or directing the model. A photograph alone does not establish understanding; no photography or public sharing is required.

Source lesson timing

Elapsed minutes	Source stage	Minutes
0–32	Independent Work	32
32–36	Lundy Loop	4
36–40	Exit Ticket	4

Split slide times are added within their original source stages. Printed tasks and adaptations replace activity within these timings; they do not add extra lesson time.

Equipment and preparation

- Original paper food/job cards and fictional case cards
- Pupil record, pencil or adult scribe
- Movable group labels; adult can read all cards aloud



TEACHING SEQUENCE

Guide the lesson

Use the source stage times; choose prompts appropriate to the pupil.

Slide 1: Compare and revise model plates · 0 minutes

Wednesday P2, 10:10-10:50. Reference screen only; start the 32-minute workshop immediately. No additional starter or lesson time.

Slide 2: Return to your labelled groups · 4 minutes

Four-minute source preparation. Use eight reference food cards and five broad group labels. Oils/spreads group remains in the reference although the eight-card set has no oil card. This is a simplified model, not actual meals.

Slide 3: Case A: group the four foods · 5 minutes

First five minutes of 20-minute investigation. Read case A as fictional food cards. Expected groups: starchy, protein, fruit/vegetables, dairy/alternatives. Ask pupil to place labels. Do not claim a complete balanced real meal from these four cards or infer portions.

Slide 4: Case B: different cards · 5 minutes

Next five minutes. Same broad group pattern as case A using different foods. Fortified soya is a labelled example, not every plant drink automatically nutritionally equivalent. Keep examples fictional and neutral.

Slide 5: Compare the two models · 5 minutes

Next five minutes. These arrows indicate a card swap, not food-chain energy arrows. Adult can read full fortified-soya-alternative label. Same group pattern does not mean matching nutrients/portions. Ask learners to explain one correspondence.



Guide the lesson - continued

Keep the source stage times; pause and check understanding as needed.

Slide 6: Make a protein-source swap · 5 minutes

Final five minutes of source 20-minute phase. Beans/lentils are protein-source examples. Lentils can be written as an additional answer; main eight-card set stays intact. Multiple alternatives allowed; no personal belief/diet justification required.

Slide 7: Pause and check what is missing · 2 minutes

Two-minute source check. Eight-card examples omit oils/spreads. A simplified single model does not show the full day/week or all requirements. Do not mark pupil as unhealthy because a paper model is incomplete.

Slide 8: Revise, explain and keep evidence · 6 minutes

Final six-minute phase includes tidy. Explanation: beans replace chicken within broad protein group; whole diets and exact amounts not established. Pupil can point to matching group or dictate a reason. No real food, photos or public sharing needed.

Slide 9: Your voice can change the lesson · 4 minutes

Four minutes within the source lesson. Receive an actual observation or access request; do not force a peer audience. Name what can change now, make the change where possible and check whether it helped. The adult receives and names back the science shown as well as access feedback.

Slide 10: What stays the same after a swap? · 4 minutes

Teacher answers only: chicken/beans/lentils/fish/eggs etc in protein group; swap preserves broad group but not exact nutrient equivalence; personal needs, full day/week intake and portions lie outside this simple model. Adult receives actual reasoning and chooses next step.



TEACHER ANSWERS

Worked answers and expected evidence

Use after an attempt. Accept equivalent correct explanations; record support given.

1. Compare the cases

Cases use different examples within same four broad groups; oils/spreads omitted from simplified eight-card activity. That limitation must be explained rather than concealed.

2. Make and justify a swap

Replace chicken with beans; lentils another valid protein source. Broad group relationship is assessed, not exact nutrient/portion equivalence. Do not require personal dietary choice explanation.

3. What can this model show?

Oils/spreads absent. Model shows broad groups and substitutions; cannot show exact portions, individual needs, complete intake over time or every mixed dish. Each meal need not replicate a diagram.

4. Improve the explanation

Credit precise nutrient/group reason and explicit limitation. Chicken and beans share broad protein group; balance concerns day/week and individual needs lie beyond cards. Adult listens to access feedback without forced disclosure.

Misconceptions and corrections

- Same food group means identical nutrient amounts: it does not.
- A meat-free choice removes the entire protein group: pulses are protein sources.
- Missing card proves an actual diet problem: fictional model limits do not diagnose pupils.



Teach safely; keep the evidence

The source record identifies this conversion; it is not a claim of publication.

Safety

- Use paper cards and invented examples only; no tasting, feeding animals or handling real food.
- Do not ask about pupils' weight, calories, personal meals or food available at home.
- Use neutral food language; no foods, bodies or families labelled good or bad.
- This is a scientific teaching model, not a diet plan; adults follow existing individual support plans.

Access and offline alternatives

- Use adult-read labels to compare cases and direct one swap.
- Name group similarities and explain one meat-free protein swap.
- Compare alternative swaps and explain the missing-group/time/portion limits.
- Adult receives one real pupil observation or access request, acts on a feasible change and explains limits. A learner can pass privately.

Close and continue

Finish at 40 minutes: workshop 32, pupil feedback 4, exit 4. Record one next teaching step based on the actual response.

Source and revision

Repository: [MattRoper1977/Lessons](https://github.com/MattRoper1977/Lessons)

Source path: [Science_Teesside/Build/SCI_B_W6_Balanced_Plate.html](#)

Source file Git blob: [186a497288fca4483f58539306d07e5a12d7d819](https://github.com/MattRoper1977/Lessons/blob/main/Science_Teesside/Build/SCI_B_W6_Balanced_Plate.html)

Current source link (mutable): https://github.com/MattRoper1977/Lessons/blob/main/Science_Teesside/Build/SCI_B_W6_Balanced_Plate.html

Document revision: 2026-09-06

Uploaded supporting sources

- White Rose Education, Year 3 Autumn Block 3 (2024): Nutrition and diet, steps 1-4, including Balanced diets and Compare diets. Original teaching models follow current website safeguards and NHS Eatwell grouping; no publisher slide/media copied.

Recorded conversion changes

- Original editable classroom text, scientific diagrams and pupil tasks follow the current weekly HTML, not the older retained A/B teaching version.
- White Rose uploads inform concepts and task choices; no publisher-owned slide, photograph, worksheet page or video is redistributed in these public files.
- Website animations and drag-and-drop controls become teacher-led reveals, movable paper labels and accessible spoken/drawn responses. No internet connection or video is needed.
- Autumn 1 Weeks 1-2 were baseline assessment. This is a Weeks 3-7 teaching pack, not a claim that baseline assessments or the separate Week 8 enrichment are included.